

Free Spectral Range Stability under Linear Thermal Tuning in a Dual-Coupled Microring Resonator

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Abstract—We demonstrate linear thermo-optical tuning of a dual-coupled microring with less than 0.2% free spectral range variation across nine modes. This regime enables thermally robust, spectrally stable operation for frequency-bin photonics, frequency combs, and co-packaged optical systems.

Index Terms—Microring resonators, free spectral range, thin-film lithium niobate, integrated photonics, optical frequency combs, dual-coupled resonators.

I. INTRODUCTION

Microring resonators play a crucial and versatile role in integrated photonics, serving as optical filters, modulators, frequency-comb generators [1], and sources for non-linear optics [2], [3]. However, the high temperature, as well as the temperature gradients that such components face when deployed in the field, causes thermal drift and mode splitting, thereby limiting their performance. To address these limitations, dual-coupled microring (DCM) resonators have been proposed, in which two coupled rings enable enhanced control over resonance behavior and mode interaction.

This work reports on the thermal and spectral stability of DCMs [4], which mitigate free spectral range (FSR) drift compared to that of a single-ring. The design was implemented in Thin-film Lithium Niobate (TFLN), which has emerged as a promising platform due to its strong electro-optic effects, favorable thermo-optic (TO) properties, and compatibility with silicon photonics [5], [6].

Previous works on DCMs analyzed mode splitting and Vernier tuning [4], [7], but not the simultaneous thermal evolution of all modes. To our knowledge, this is the first study to characterize the multi-mode FSR stability of a TFLN DCM under linear thermal tuning.

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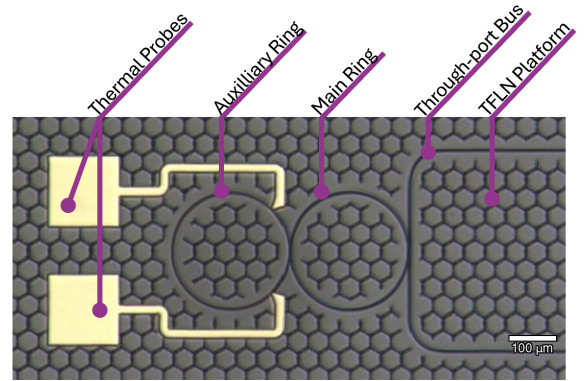


Fig. 1. Microscopic image of the TFLN DCM. The main and auxiliary rings (150 μm and 149 μm radius) are evanescently coupled via 300 nm gaps and integrated with resistive thermal probes for controlled tuning.

Our measurements demonstrate sub 0.2% FSR variation and highly linear tuning ($R^2 \approx 0.94$), underscoring the potential of TFLN DCMs for stable filters and wavelength-tunable photonic systems.

II. EXPERIMENTAL RESULTS

Our DCM, shown in Fig. 1, consists of two rings, a main ring of radius 150 μm and an auxiliary ring of radius 149 μm . The slight mismatch in radii imposes a Vernier offset between the rings, enabling a degree of freedom to selectively enhance mode coupling at wavelengths where the two spectra locally align. The main ring is coupled to the throughport by a gap of 300 nm and to the auxiliary ring by a gap of 300 nm. The TFLN has a group index of 2.3003, and supports single-mode TE operation in the C-band, at $\lambda \approx 1550$ nm. Thermal tuning can be applied to the device via integrated resistive probes connected to the auxiliary ring.

For our experiment, we measured the transmission of the device under a broadband input signal using a high-resolution

optical spectrum analyzer, while linearly sweeping the heater current from 0 to 30 mA in 31 steps.

The measured FSR is 1.10 ± 0.002 nm (~ 140 GHz), within 1% of its theoretical value:

$$\text{FSR} = \frac{\lambda^2}{n_g L}, \quad (1)$$

where λ denotes the resonance wavelength, n_g is the group index, and $L = 2\pi R$ is the round-trip optical path length corresponding to the radius R of the main ring.

The stability trend in Fig. 2(a) shows that throughout the full tuning range, the FSR changes less than 0.2%, demonstrating exceptional spectral stability for a thermally tuned resonator.

Fig. 2(b) shows that the resonances red-shift linearly with heater power at approximately 2 nm/W, with a fitting correlation of $R^2 \approx 0.94$. All modes follow the same slope, and their line-widths and Q-factors remained consistent, confirming a uniform thermal response and a stable coupling regime.

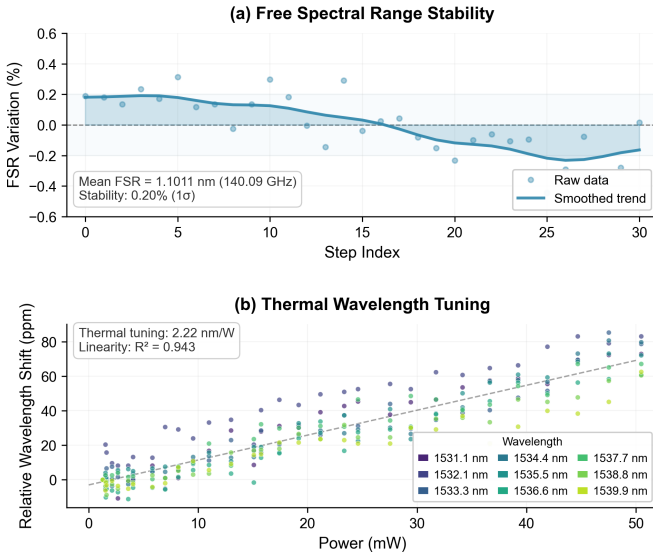


Fig. 2. (a) FSR stability across 31 thermal control steps. Raw data points (light blue) show a 1.1011 nm (140.09 GHz) variation from the mean. The dark blue line shows a Gaussian-smoothed trend ($\sigma = 2$ steps). FSR stability is 0.20% (1σ standard deviation, 1964 ppm). (b) Thermal wavelength tuning for nine cavity modes spanning 1531–1540 nm. A linear fit (dashed gray) demonstrates tuning efficiency of 2.22 nm/W with $R^2 = 0.943$.

III. DISCUSSION

In TFLN, the TO coefficient $\frac{dn}{dT} \approx 3.6 \times 10^{-5}/\text{K}$ and the thermal-expansion coefficient $\alpha \approx 1.5 \times 10^{-5}/\text{K}$ are positive [6]. Consequently, each resonance red-shifts with increasing temperature, while the FSR decreases only slightly. Both n_g and L increase with temperature, and their fractional changes add as follows:

$$\frac{\Delta \text{FSR}}{\text{FSR}} \approx - \left(\frac{\Delta n_g}{n_g} + \frac{\Delta L}{L} \right) \approx -3.1 \times 10^{-5} \text{ K}^{-1}. \quad (2)$$

This intrinsic balance between refractive-index and geometric expansion explains much of the observed $< 0.2\%$ variation.

The slightly mismatched radii of the DCM's further enhance stability, as it leads to comparable slopes $\frac{dn}{dT}$, allowing the coupled modes to shift coherently. Although resonant doublets were not tracked in this experiment, such a configuration inherently supports Vernier alignment preservation and an extended tuning range without loss of coupling [7].

These results highlight that the combination of TFLN material properties and the coupled-ring geometry effectively suppresses FSR drift under thermal tuning.

IV. CONCLUSION

We demonstrate linear thermo-optic tuning in a dual-coupled TFLN microring resonator with exceptional FSR stability ($< 0.2\%$ variation) across a 2 nm/W tuning range and $R^2 \approx 0.94$ linearity. This stability arises from the favorable material properties of TFLN. Small TO and thermal-expansion coefficients yield a fractional FSR drift of only $-3.1 \times 10^{-5} \text{ K}^{-1}$.

The demonstrated FSR-invariant tuning regime has important implications for thermally sensitive applications. In co-packaged optics, where electronic chips generate substantial on-chip thermal gradients, maintaining stable spectral grids is critical for wavelength-division multiplexing and optical interconnects. Similarly, frequency-bin quantum photonics and microresonator frequency combs require precise FSR control to preserve mode relationships and spectral purity. Our results confirm that TFLN dual-coupled microrings can operate reliably in these thermally dynamic environments while maintaining predictable wavelength control.

This work validates a simple, passive approach to thermally robust integrated photonics, enabling predictable wavelength control in compact, power-efficient systems where thermal variations are unavoidable.

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